



LIGHT BEAMS AND JOISTS

Section Size	Moment of inertia				Radius of gyration				Modulus of section			
	Ix		Iy		ix		iy		Zx		Zy	
	in ⁴	cm ⁴	in ⁴	cm ⁴	in	cm	in	cm	in ³	cm ³	in ³	cm ³
6 x 4 (J)	14.8	616	1.89	78.66	2.43	6.17	0.87	2.21	5.07	83.1	0.96	15.73
6 x 4 (L)	21.7	903.3	2.89	120.3	2.48	6.30	0.90	2.29	7.24	118.7	1.44	23.60
	31.7	1,320	4.32	179.8	2.59	6.58	0.96	2.44	10.1	165.5	2.14	35.07
8 x 4 (J)	30.8	1,282	1.99	82.82	3.23	8.20	0.82	2.08	7.79	127.7	1.01	16.55
8 x 4 (L)	39.5	1,644	2.62	109.1	3.21	8.15	0.83	2.11	9.88	161.9	1.31	21.47
	48.0	1,994	3.30	137.4	3.29	8.36	0.86	2.18	11.8	193.4	1.65	27.04
10 x 4 (J)	51.9	2,160	2.01	83.66	3.92	9.96	0.77	1.96	10.5	172.1	1.02	16.72
10 x 4 (L)	68.8	2,864	2.79	116.1	3.95	10.03	0.80	2.03	13.8	226.2	1.39	22.78
	81.8	3,405	3.45	143.6	4.05	10.29	0.83	2.11	16.2	265.5	1.72	28.19
	96.2	4,005	4.19	174.4	4.14	10.52	0.86	2.18	18.8	308.1	2.08	34.09
12 x 4 (J)	88.2	3,671	2.25	93.65	4.61	11.71	0.74	1.88	14.8	242.5	1.13	18.52
12 x 4 (L)	105.3	4,383	2.79	116.1	4.65	11.81	0.76	1.93	17.5	286.8	1.39	22.78
	130.1	5,416	3.67	152.8	4.81	12.22	0.81	2.06	21.4	350.7	1.83	29.99
	155.7	6,482	4.55	189.4	4.91	12.47	0.84	2.13	25.3	414.7	2.26	37.04
14 x 4 (L)	147.3*	6,131*	2.65*	110.3 *	5.40*	13.72*	0.72*	1.83*	21.0 *	344.2*	1.32*	21.63*
14 x 5 (L)	197.4	8,212	6.40	266.4	5.52	14.02	0.99	2.51	28.8	472.0	2.56	41.96
	242.6	10,100	8.26	343.8	5.63	14.30	1.04	2.64	34.9	572.0	3.29	53.92
16 x 5 1/2 (L)	298.1	12,410	8.71	362.5	6.24	15.85	1.07	2.72	38.1	624.5	3.17	51.96
	372.5	15,500	11.57	481.5	6.39	16.23	1.13	2.87	47.0	770.3	4.19	68.67